MRI Full Course Formula Sheet

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Week 1 - NMR Basics

Spin

Angular momentum	$p = mv$, $J = r \times p$	Rotating charge.
Magnetic moment	$\mu = \gamma J$	γ gyromagnetic ratio.
Quantization	$J_z = \hbar m$, $m = I, I-1, \ldots, -I$	Spin projection.
Spin 1/2	$I = 1/2 \Rightarrow m = \pm 1/2$	^{1}H has two states.
Moment projection	$\mu_z = \gamma\hbar m = \pm(1/2)\gamma\hbar$	Along B_0.

Energy And Resonance

Energy	$E_m = -\mu_z B_0 = -\gamma\hbar B_0 m$	Parallel lower for $\gamma>0$.
Gap	$\Delta E = \gamma\hbar B_0 = \hbar\omega_0$	Energy splitting.
Boltzmann	$n_{down}/n_{up} = \exp(-\Delta E/k_B T)$	Population ratio.
Polarization	$\Delta n/n \approx \Delta E/(2k_B T)$	For $\Delta E \ll kBT$.
Net M	$M_0 = \sum\mu = \Delta n\ \mu_z e_z$	Longitudinal M.
Scaling	$M_0 \propto \rho_H B_0/T$	Proton density matters.

Dynamics

Torque	$d\mu/dt = \gamma(\mu \times B)$	Precession.
Macroscopic	$dM/dt = \gamma(M \times B)$	Same law for M.
RF	$B_1(t)=B_1\cos(\omega_0 t)e_x=B_L+B_R$	Excitation field.
Rotating frame	$(dM/dt)_{rot}=\gamma M\times B-\omega_0\times M=\gamma M\times B_{eff}$	Use effective field.
Effective field	$B_{eff}=(B_0-\omega/\gamma)e_{z'}+B_{1,rot}e_{x'}$	On resonance mostly B1.
Flip angle	$\alpha = \gamma\int B_{1,rot}(t)dt$	RF pulse area.

Relaxation

T2*	$1/T_2^* = 1/T_2 + \gamma\delta B$	Field inhomogeneity.
Bloch x	$dM_x/dt=\gamma(M_y B_z-M_z B_y) - M_x/T_2$	Transverse loss.
Bloch y	$dM_y/dt=\gamma(M_z B_x-M_x B_z) - M_y/T_2$	Transverse loss.
Bloch z	$dM_z/dt=\gamma(M_x B_y-M_y B_x) - (M_z-M_0)/T_1$	Recovery.

Week 2 - Image Formation

Encoding

Gradient	$G_i=\partial B_z/\partial i$	i=x,y,z.
Loop field	$B_z(z)=\mu_0 I\ r^2/(2(r^2+z^2)^{3/2})$	Circular current loop.
Linear field	$\Delta B_z(x)=G_x\ x$	Constant slope.
Frequency	$\omega(r)=\gamma(B_0+G\ r)$	Position dependent.
Readout	$\Delta\omega(x)=\gamma G_x\ x$	Frequency encoding.
Phase	$\varphi(y)=\gamma G_y\ y\ T_y$	Phase encoding.
K-space	$k(t)=\gamma\int G(t)dt$	Constant G: $k=\gamma Gt$.

Signal And FT

Signal	$s(k_x,k_y) \propto \sum_{i,j} \rho(x_i,y_j) \exp[j(k_x x_i+k_y y_j)]$	FT of object.
Image	$\rho(x,y) \propto \sum_{p,q} s(k_{x,p},k_{y,q}) \exp[-j(k_{x,p}x+k_{y,q}y)]$	Inverse FT.
K spacing	$\Delta k = 2\pi/FOV = BW / N_x$	Nyquist/readout spacing.
Pixel	$\Delta x=FOV_x/N_x$	Same in y.
Max k	$k_{max}=\pi/\Delta x=\pi N_x/FOV_x$	Resolution.
Readout BW	$BW_x=\gamma G_x FOV_x$	Angular freq convention.
Dwell	$\Delta t=2\pi/BW_x$, $T_x=N_x\Delta t$	Sampling.

Slice And Contrast

Gradient moment	$k_i=\gamma\int G_i(t)dt$	Gradient area.
Slice thickness	$\Delta z=BW_{RF}/(\gamma G_z)$	RF bandwidth.
RF selectivity	$R=BW_{RF}T_{RF}$	Longer pulse narrower BW.
PD contrast	$C\propto\rho_A-\rho_B$	Long TR.
Partial sat.	$C\propto\rho_A[1-\exp(-TR/T_{1A})]-\rho_B[1-\exp(-TR/T_{1B})]$	T1 weighting.
Scan time	$T_{scan}\approx N_{phase}TR$	One ky line per TR.

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Week 3 - Fast And Parallel Imaging

Speed, SNR, Resolution

Voxel	$\Delta V = \Delta x \Delta y \Delta z$	Voxel volume.
SNR scaling	$SNR \propto \Delta V \sqrt{T_{scan}} / \sqrt{BW}$	Core tradeoff.
Averages	$SNR \propto \sqrt{NSA}$	Repeated scans.
Acceleration	$T_{scan,R} \approx T_{scan}/R$	Undersampling.
Aliased FOV	$FOV_{alias} = FOV/R$	Phase undersampling.

Coil Encoding

Coil signal	$s_\gamma(k) = \int \rho(x) c_\gamma(x) \exp(ikx)\, dx$	Sensitivity cγ.
Matrix model	$s = E\rho + \eta$	Encoding plus noise.
Decode	$i = Fs$, $F E = I$	Ideal reconstruction gives identity response.
SENSE	$\hat{\rho} = (E^H \Psi^{-1} E)^{-1} E^H \Psi^{-1} s$	Optimum SNR inverse.
Regularized	$\hat{\rho} = (E^H \Psi^{-1} E + \lambda I)^{-1} E^H \Psi^{-1} s$	Stabilizes ill-conditioning.

Parallel Limits

SENSE SNR	$SNR_{SENSE} = SNR_{full} / (\sqrt{R} \cdot g(x))$	g-factor penalty.
g-factor	$g(x) = \sqrt{[(E^H E)_{ii} ((E^H E)^{-1})_{ii}]} \geq 1$	Noise amplification.
Good coils	Distinct $c_\gamma(x) \Rightarrow$ low g	Separates aliased voxels.
Failure	$R > N_{coils}$ or poor sensitivities	Underdetermined/ill-conditioned.

Week 4 - Image Contrast

Relaxation Contrast

T1 recovery	$M_z(t) = M_0 + [M_z(0) - M_0] \exp(-t/T_1)$	Longitudinal.
After 90°	$M_z(t) = M_0[1 - \exp(-t/T_1)]$	Mz(0)=0.
T2 decay	$M_{xy}(t) = M_{xy}(0) \exp(-t/T_2)$	Spin-spin.
T2* decay	$M_{xy}(t) = M_{xy}(0) \exp(-t/T_2^*)$	GRE/FID.

Common Signals

Spin echo	$S_{SE} \propto \rho [1 - \exp(-TR/T_1)] \exp(-TE/T_2)$	T1 by TR, T2 by TE.
Max contrast TR	$T_R = [T_{1A} T_{1B} / (T_{1B} - T_{1A})] \ln[(S_A T_{1B})/(S_B T_{1A})]$	Obtained by setting dC/dTR=0.
GRE	$S_{GRE} \propto \rho \sin\alpha\, (1 - E_1)/(1 - E_1 \cos\alpha) \exp(-TE/T_2^*)$	E1=exp(-TR/T1).
Inversion	$M_z(TI) = M_0[1 - 2\exp(-TI/T_1)]$	TR >> T1.
Null time	$TI_{null} = T_1 \ln 2$	Suppress tissue.
Ernst	$\alpha_E = \arccos[\exp(-TR/T_1)]$	Max steady-state signal.

Weighting Rules

PD	TR long, TE short	Minimize relaxation weighting.
T1	TR short, TE short	T1 recovery dominates.
T2	TR long, TE long	T2 decay dominates.
T2*	GRE with TE long	Sensitive to dephasing.

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Week 5 - SNR And Hardware

Signal And Noise

Signal voltage	$U_{sig}(x)=\omega M_{xy}(x)C(x)\Delta V$	Receive sensitivity C.
Noise power	$P=\sigma\|E\|^2\Delta V$	Sample losses.
Noise variance	$\Psi=4k_B T\ BW\ R$	Johnson-Nyquist.
SNR	$SNR=U_{sig}/U_{noise}$	Image quality.
SNR scaling	$SNR\propto\Delta V\sqrt{NSA}/\sqrt{BW}$	Resolution/speed tradeoff.

Optimization

Increase signal	$\uparrow C,\ \uparrow M_{xy},\ \uparrow\omega,\ \uparrow\Delta V$	More voltage.
Reduce noise	$\downarrow BW,\ \downarrow T,\ \downarrow R_{coil}$	Less thermal noise.
Field scaling	$M_0\propto B_0,\ \omega\propto B_0$	U_{sig} roughly grows strongly with B0.
Surface coil	C(x) high near coil	High local SNR.

Resolution Limits

Fourier	$\Delta x\approx\pi/k_{max}$	Sampling aperture.
Relaxation blur	$k^* = \gamma G T_2^*$	Finite T2* filters k-space.
Diffusion blur	$x^2\approx 6DT_{acq}$	Long readouts blur.
SoS combine	$I_{SoS}=\sqrt{(\sum_c\|I_c\|^2)}$	Magnitude coil combine.

Week 6 - Flow Imaging

Motion Phase

Trajectory	$r(t)=r_0+v(t-t_0)+a(t-t_0)^2/2+\ldots$	Moving spin.
Phase	$\varphi=\gamma\int G(t)r(t)dt$	Gradient phase.
Moments	$M_n=\int G(t)(t-t_0)^n dt$	nth gradient moment.
Expansion	$\varphi=\gamma[r_0M_0+vM_1+aM_2/2+\ldots]$	Position/velocity/acceleration.
Bipolar	$M_0=0,\ \varphi=\gamma vM_1$	Velocity encoding.

Phase Contrast

Encoding	$VENC=\pi/(\gamma M_1)$	Phase reaches ±π.
Velocity	$v=(\Delta\varphi/\pi)VENC$	Phase difference map.
Aliasing	$\|v\|>VENC \Rightarrow$ phase wraps	Set VENC high enough.
Flow rate	$Q=\sum_i v_i\Delta A_i$	Through-plane flow.
Velocity distribution	$s(x,k_v)=\sum\rho(x,v)\exp(jk_v v)$	Generalized velocity encoding.

Hemodynamics

Poiseuille	$Q=\Delta P/R,\ R\propto 1/d^4$	Diameter dominates.
Reynolds	$Re=\rho vd/\mu$	Laminar if Re<~2000.
Shear stress	$SS=-\mu\ \partial v/\partial r$	Wall shear.
Contrast agent	$1/T_{1,app}=1/T_1+R_1c$	Relaxivity model.

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Week 7 - Motion And Artifacts

Motion Encoding

Phase	$\varphi=\gamma\int G(t)x(t)dt$	All motion artifacts start here.
Motion model	$x(t)=x_0+v(t-t_0)+a(t-t_0)^2/2+...$	Taylor expansion.
Moment nulling	$M_0=0$ removes position phase; $M_1=0$ compensates velocity	Gradient design.
Velocity phase	$\varphi_v=\gamma v M_1$	Residual if M1 nonzero.
Velocity spread	signal ∝ sinc(βγAΔt Δy/2)	Intravoxel dephasing.

Artifact Rules

Intra-TR	motion during readout/echo ⇒ phase errors and signal loss	Flow, pulsation.
Inter-TR	motion between ky lines ⇒ ghosting/blurring	Respiration/patient.
Periodic ghosts	ghost spacing in phase direction ∝ $1/(f_{motion}TR)$	Regular motion.
Random motion	random ky phase ⇒ diffuse blur	Irregular motion.

Correction

Rigid transform	$R(r,t)=A(t)r+d(t)$	Rotation plus translation.
Prospective gradients	$G'(t)=A(t)^{-T}G(t)$	Follow anatomy.
Phase correction	$\varphi'(t)=\varphi(t)-\gamma G(t)\cdot d(t)$	Translation compensation.

Week 8 - fMRI And Diffusion

BOLD/fMRI

T2*	$1/T_2^*=1/T_2+\gamma\Delta B$	Static dephasing.
Susceptibility	$\Delta B\approx\Delta\chi B_0$	Blood oxygenation effect.
GRE BOLD	$S(TE)=S_0\exp(-TE/T_2^*)$	TE near T2*.
Small change	$\Delta S/S\approx-TE\cdot\Delta R_2^*$	R2*=1/T2*.

Diffusion Physics

Fick	$j=-D\nabla c(r,t)$	Diffusion flux.
Diffusion eq.	$\partial c/\partial t=D\nabla^2 c$	Conservation plus Fick.
RMS displacement	$R_{rms}=\sqrt{(6D\Delta)}$	3D free diffusion.
Einstein	$D\approx v^2\tau/6$	Random walk.

DWI/DTI

DWI signal	$S(TE,b)=S_0\exp(-TE/T_2)\exp(-bD)$	Scalar diffusion.
b-value	$b=\gamma^2G^2\delta^2(\Delta-\delta/3)$	PGSE sensitivity.
Tensor	$\ln(S/S_0)=-b\, g^TDg$	Direction g.
MD	$MD=(\lambda_1+\lambda_2+\lambda_3)/3$	Mean diffusivity.
FA	$FA=\sqrt{(3/2)}\,\sqrt{\sum(\lambda_i-MD)^2}/\sqrt{\sum\lambda_i^2}$	Anisotropy.

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Week 9 - Advanced Imaging

General Encoding

Data model	$d=E\rho+\eta$	General MRI inverse problem.
Cartesian FT	E=Fourier sampling operator	Standard MRI.
Parallel	E includes coil sensitivities $c_\gamma(r)$	SENSE/arrays.
Optimal inverse	$\hat\rho=(E^H\Psi^{-1}E)^{-1}E^H\Psi^{-1}d$	If well-conditioned.
Regularized	$\hat\rho=(E^H\Psi^{-1}E+\lambda I)^{-1}E^H\Psi^{-1}d$	For high R.

Compressed Sensing

Sparsity	$x=\Phi\rho$ has many small/zero coefficients	Sparse transform.
Undersampled data	$d_\Omega=P_\Omega F\rho$	Sample subset Ω.
CS recon	$\min_\rho \|\|\Phi\rho\|\|_1$ s.t. $\|\|E\rho-d\|\|_2\leq\varepsilon$	Sparse recovery.
TV	$\Phi\rho=\nabla\rho \Rightarrow$ total variation penalty	Piecewise smooth images.
Sampling	random/incoherent k-space undersampling	Artifacts become noise-like.

Limits

PSF	$PSF(r)=FT^{-1}\{\text{sampling pattern}\}$	Undersampling artifact shape.
Acceleration	higher R needs sparsity + SNR + calibration	No free lunch.
Low-rank/dynamics	suppp small or temporal basis small	Advanced priors.

Week 10 - Spectroscopy I

Chemical Shift

Larmor	$\nu=-\gamma B_0$ [Hz]	Sign by convention.
Shift	$\delta=(\nu-\nu_{ref})/\nu_{ref}\cdot 10^6$ ppm	Field-independent ppm.
Hz separation	$\Delta\nu=\Delta\delta\cdot\nu_{ref}\cdot 10^{-6}$	Grows with B0.
1 ppm	1 ppm = $\nu_{ref}\cdot 10^{-6}$ Hz	64 Hz at 1.5T for 1H.

Spectrum

FID	$s(t)=A\exp(-t/T_2^*)\exp(i2\pi\nu_A t)$	Single component.
Spectrum	$S(\nu)=FT\{s(t)\}$	Frequency-domain signal.
Lorentzian	$\mathrm{Re}\,S(\nu) \propto T_2^* / [1+(2\pi T_2^*(\nu-\nu_A))^2]$	Line shape.
Linewidth	$FWHM=1/(\pi T_2^*)$	Shorter T2* broader.
Mixture	$s(t)=\sum_c w_c \exp(-t/T_{2,c}^*)\exp(i2\pi\nu_c t)$	Superposition.

Sampling And J

Acquisition	$T_{acq}=Ndt$	N samples.
Bandwidth	$BW=1/dt$	Hz.
Resolution	$d\nu=1/T_{acq}$	Peak spacing.
Resolve peaks	$d\nu \leq \Delta\nu/2$	Rule of thumb.
J multiplet	N coupled spin-I nuclei $\Rightarrow$ 2NI+1 lines	Binomial intensities.
J echo	TE=n/J	Odd n flips doublet sign.
Editing	TE=1/J	Difference editing.

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Week 11 - Spectroscopy II

Suppression And Inversion

Water/fat	$\Delta\delta \approx 3.5$ ppm	Water 4.7, fat 1.2 ppm.
Phase accrual	$\Delta\varphi = 2\pi\Delta\nu\tau$	Chemical shift phase.
Binomial spacing	$\tau \approx 1/(2\Delta\nu)$ or $1/(4\Delta\nu)$	Depends on target phase.
Inversion recovery	$M_z(TI) = M_0[1-2\exp(-TI/T_1)]$	After 180°.
Null	$TI_{null} = T_1 \ln 2$	Water/fat suppression.

Localization

Surface coil	$S(r)=B_1(r) \sin[\gamma B_1(r)t]$	Sensitivity + flip angle.
Adiabatic	$\lvert d\theta/dt\rvert \ll \gamma\lvert B_{eff}\rvert$	Robust inversion/excitation.
PRESS/STEAM	Echo at TE; stimulated echo stores Mz during TM	Voxel selection.
Phase cycling	linear combinations isolate voxel term	Remove unwanted echoes.

CSI And High Field

Shift artifact	$\Delta x = \Delta\nu_{CS}/(\gamma G)$	Chemical shift displacement.
Fractional shift	$\Delta x/\Delta x_{RF} = \Delta\nu_{CS}/BW_{RF}$	Use large RF BW.
CSI signal	$s(k_x,k_y,t)=\iiint \rho(x,y,\Delta\nu) \exp[i(k_x x+k_y y+2\pi\Delta\nu t)]\, dx dy d\Delta\nu$	2 spatial + spectral.
CSI recon	$S(x,y,\Delta\nu)=FT_t FT_{kx} FT_{ky}\{s\}$	3D transform.
CSI scan time	$T_{scan}=N_x N_y TR\cdot NSA$	Slow phase encoding.
High-field SNR	$U_{sig}\sim B_0^2$, $U_{noise}\sim B_0$, $SNR\sim B_0$	Approximate.
SAR	$P \propto \sigma E^2 \propto \sigma\gamma^2 B_0^2 B_1^2$	High-field cost.